Course code	Course Name	Teaching Scheme (Contact Hours)			Credits Assigned			
		Theory	Practical	Tut	Theory	Practical	Tut	Total
PCC702	Big Data Analytics	3	-	-	3	-	-	3

Course Code	Course Name	Theory					Term work	Pract / Oral	Total
		Internal Assessment			End SEM Exam	Exam duration (in hrs)			
		Test 1	Test 2	Avg					
PCC702	Big Data Analytics	20	20	20	80	3	-	-	100

Course Objectives: The Course aims to

1. To provide an overview of an exciting growing field of big data analytics.
2. To introduce programming skills to build simple solutions using big data technologies such as MapReduce and scripting for NoSQL, and the ability to write parallel algorithms for multiprocessor execution
3. To teach the fundamental techniques and principles in achieving big data analytics with scalability and streaming capability.
4. To enable students to have skills that will help them to solve complex real-world problems in decision support.
5. To provide an indication of the current research approaches that is likely to provide a basis for tomorrow's solutions.

Course Outcomes: Learners will be able to

1) Understand the key issues in big data management and its associated applications for business decisions and strategy.
2) Develop problem solving and critical thinking skills in fundamental enabling techniques like Hadoop, Map reduce and NoSQL in big data analytics.
3) Collect, manage, store, query and analyze various forms of Big Data.
4) Interpret business models and scientific computing paradigms, and apply software tools for big data analytics.
5) Adapt adequate perspectives of big data analytics in various applications like recommender systems, social media applications etc.
6) Solve Complex real world problems in various applications like recommender systems, social media applications, health and medical systems, etc.

Prerequisite: Some prior knowledge about Java programming, Basics of SQL, Data Mining and Machine Learning methods would be beneficial.

DETAILED SYLLABUS: total six module for each subject (13 Weeks)

Sr. No.	Module	Detailed Content	Hours	CO Mapping
I	**Introduction to Big Data & Hadoop**	1.1 Introduction to Big Data 1.2 Big Data characteristics, types of Big Data, 1.3 Traditional vs. Big Data business approach, 1.4 Case Study of Big Data Solutions. 1.5 Concept of Hadoop 1.6 Core Hadoop Components; Hadoop Ecosystem	**05**	**CO2**
II	**Hadoop HDFS and Map Reduce**	2.1 Distributed File Systems: Physical Organization of Compute Nodes, Large-Scale File-System Organization. 2.2 MapReduce: The Map Tasks, Grouping by Key, The Reduce Tasks, Combiners, Details of MapReduce Execution, Coping With Node Failures. 2.3 Algorithms Using MapReduce: Matrix-Vector Multiplication by MapReduce, Relational-Algebra Operations, Computing Selections by MapReduce, Computing Projections by MapReduce, Union, Intersection, and Difference by MapReduce 2.4 Hadoop Limitations.	**08**	**CO2** **CO3** **CO4**
III	**NoSQL**	3.1 Introduction to NoSQL, NoSQL Business Drivers 3.2 NoSQL Data Architecture Patterns: Key-value stores, Graph stores, Column family (Bigtable)stores, Document stores, Variations of NoSQL architectural patterns, NoSQL Case Study 3.3 NoSQL solution for big data, Understanding the types of big data problems; Analyzing big data with a shared-nothing architecture; Choosing distribution models: master-slave versus peer-to-peer; NoSQL systems to handle big data problems. peer-to-peer; Four ways that NoSQL systems handle big data problems	**05**	**CO2** **CO3**
IV	**Mining Data Streams**	4.1 The Stream Data Model: A Data-Stream-Management System, Examples of Stream Sources, Stream Queries, Issues in Stream Processing. 4.2 Sampling Data techniques in a Stream 4.3 Filtering Streams: Bloom Filter with Analysis.	**10**	**CO4** **CO5**

		4.4 Counting Distinct Elements in a Stream, Count-Distinct Problem, Flajolet-Martin Algorithm, Combining Estimates, Space Requirements 4.5 Counting Frequent Items in a Stream, Sampling Methods for Streams, Frequent Itemsets in Decaying Windows. 4.6 Counting Ones in a Window: The Cost of Exact Counts, The Datar-Gionis-Indyk-Motwani Algorithm, Query Answering in the DGIM Algorithm, Decaying Windows.		
V	**Finding Similar Items and Clustering**	5.1 Distance Measures: Definition of a Distance Measure, Euclidean Distances, Jaccard Distance, Cosine Distance, Edit Distance, Hamming Distance 5.2 CURE Algorithm, Stream-Computing, A Stream-Clustering Algorithm, Initializing & Merging Buckets, Answering Queries.	**06**	**CO5 CO6**
VI	**Real-Time Big Data Models**	6.1 PageRank Overview, Efficient computation of PageRank: PageRank Iteration Using MapReduce, Use of Combiners to Consolidate the Result Vector. 6.2 A Model for Recommendation Systems, Content-Based Recommendations, Collaborative Filtering 6.3 Social Networks as Graphs, Clustering of Social-Network Graphs, Direct Discovery of Communities in a social graph.	**08**	**CO5 CO6**
		Total No. of Lectures:	**42**	

Text Books:

1. Anand Rajaraman and Jeff Ullman ―Mining of Massive Datasets‖, Cambridge University Press,
2. Alex Holmes ―Hadoop in Practice‖, Manning Press, Dreamtech Press.
3. Dan Mcary and Ann Kelly ―Making Sense of NoSQL‖ – A guide for managers and the rest of us, Manning Press.

References:

1. Bill Franks, ―Taming The Big Data Tidal Wave: Finding Opportunities In Huge Data Streams With Advanced Analytics‖, Wiley
2. Chuck Lam, ―Hadoop in Action‖, Dreamtech Press
3. Jared Dean, ―Big Data, Data Mining, and Machine Learning: Value Creation for Business Leaders and Practitioners‖, Wiley India Private Limited, 2014.
4. Jiawei Han and Micheline Kamber, ―Data Mining: Concepts and Techniques‖, Morgan Kaufmann Publishers, 3rd ed, 2010.

5. Lior Rokach and Oded Maimon, ―Data Mining and Knowledge Discovery Handbook‖, Springer, 2nd edition, 2010.
6. Ronen Feldman and James Sanger, ―The Text Mining Handbook: Advanced Approaches in Analyzing Unstructured Data‖, Cambridge University Press, 2006.
7. Vojislav Kecman, ―Learning and Soft Computing‖, MIT Press, 2010

Online References:

Sr. No.	Website Name
1.	https://nptel.ac.in/courses/106104189
2.	https://nptel.ac.in/courses/106106169

Assessment:

Internal Assessment (IA) for 20 marks:

- ✓ IA will consist of Two Compulsory Internal Assessment Tests. Approximately 40% to 50% of syllabus content must be covered in First IA Test and remaining 40% to 50% of syllabus content must be covered in Second IA Test.

Question paper format:

- Question Paper will comprise of a total of **six questions each carrying 20 marks Q.1** will be **compulsory** and should **cover maximum contents of the syllabus.**
- From the **remaining five questions, attempt any three. The Remaining five questions** will be **mixed in nature** (part (a) and part (b) of each question must be from different modules. For example, if Q.2 has part (a) from Module 3 then part (b) must be from any other Module randomly selected from all the modules).
- A total of **Four questions** need to be answered.